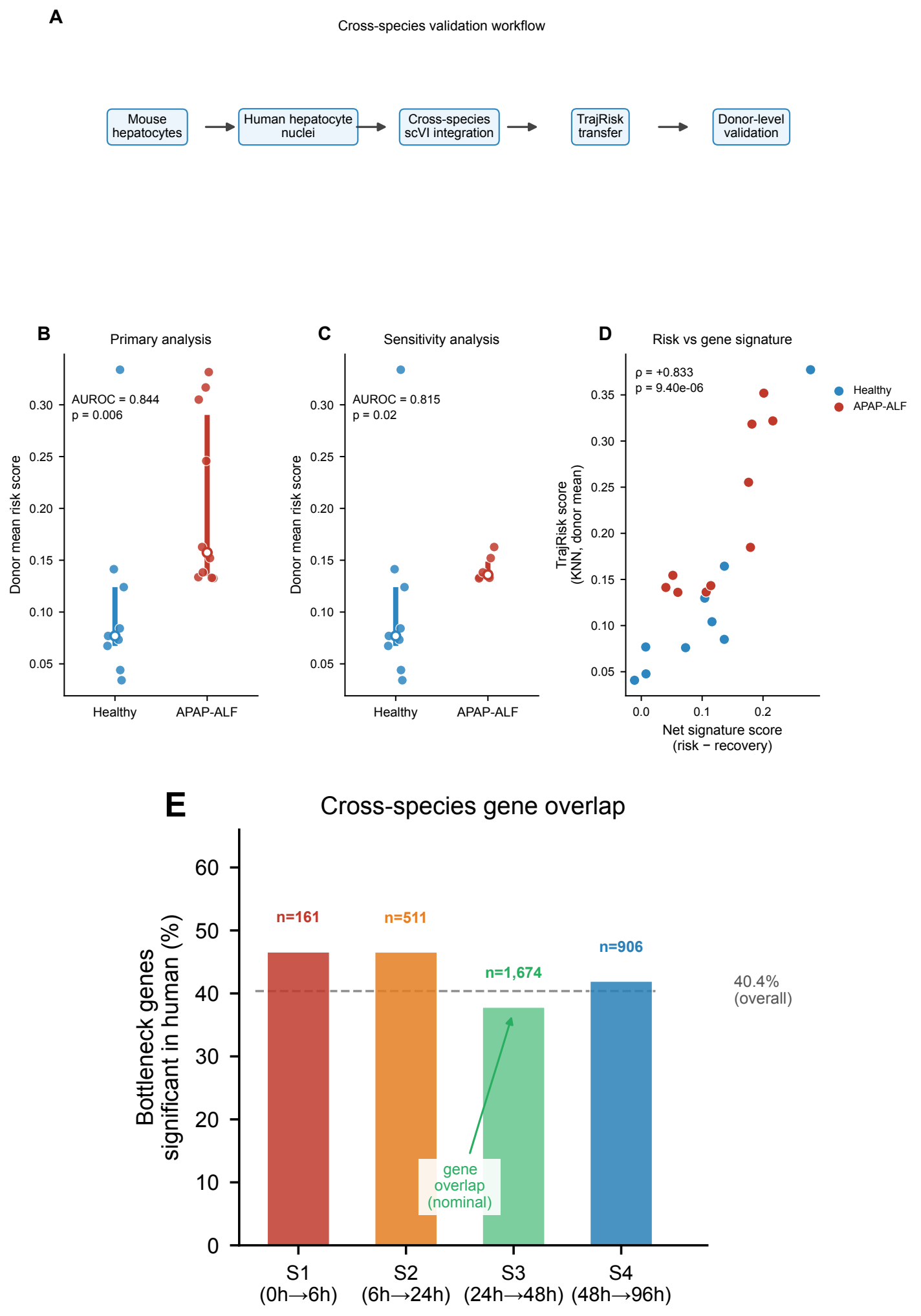

A
Cross-species validation workflow
Mouse hepatocytes
Human hepatocyte nuclei
Cross-species scVI integration
TrajRisk transfer
Donor-level validation
B
Primary analysis
AUROC = 0.844
p = 0.006
Donor mean risk score
Healthy
APAP-ALF
C
Sensitivity analysis
AUROC = 0.815
p = 0.02
Donor mean risk score
Healthy
APAP-ALF
D
Risk vs gene signature
ρ = +0.833
p = 9.40e-06
TrajRisk score (KNN, donor mean)
Net signature score (risk − recovery)
Healthy
APAP-ALF
E
Cross-species gene overlap
Bottleneck genes significant in human (%)
n=161
n=511
n=1,674
n=906
40.4% (overall)
gene overlap (nominal)
S1 (0h→6h)
S2 (6h→24h)
S3 (24h→48h)
S4 (48h→96h)